Résumé × JD Analyser — Lab Guide

Note to the Reader

> This lab is the practical companion to the study material. You are not starting from scratch — you have a starter scaffold with the boilerplate already in place. Your job is to build the four files that contain the conceptual weight: `parse.py` , `prompts.py` , `analyzer.py` , and `main.py` .
>
> Every task ends with a **Checkpoint** — a runnable command that confirms only that layer works before you continue. You should never be building blind.
>
> **Track B approach:** You will describe what you want in plain English, ask Claude to generate the Python, and verify each result with the Checkpoint. This is not a simplified version of the lab — it is a different skill. Describing a system accurately enough for an AI to implement it *requires* understanding the system. The study material gives you that understanding; this lab is where you apply it.

What You'll Build

This lab produces a Python command-line tool: the **Résumé × JD Analyser** — a small-scale model of the applicant-screening pipeline a company's recruiting team might run. You supply a candidate's PDF résumé and a job description; the tool runs a multi-stage LLM pipeline and produces a structured screening report with a 0–100 score against a 60% pass threshold, indicating whether the résumé merits advancing to interview. The tool is **feedback-only** — its job is evaluation, not ghostwriting: it scores and explains, but it never rewrites the candidate's résumé.

The four pipeline stages:

Step through what a complete successful run looks like:

Next Message | Play All | Replay | 0 / 5 messages

Companion Reading

Keep the study material open alongside this tutorial. Each task links to the relevant section.

Task	Study material section
Task 1: `parse.py`	§5 Document Parsing
Task 2: `llm.py`	§8 Handling Imperfect AI Output
Task 3: `prompts.py`	§3.3 Schema-First Prompt Design · §6.1 Extraction Prompts · §6.2 Evaluation Prompts · §6.3 Feedback-Only Principle
Task 4: `analyzer.py`	§4 The Multi-Stage Pipeline · §7.2 Weighted Aggregation
Task 5: `main.py`	§4.3 Data Flow Between Stages

Prerequisites

- Comfort with LLM completion calls (e.g. via LiteLLM or an equivalent SDK), the `messages` array, `json.loads` / `json.dumps`, `.env` secrets management, and the ICCO prompt-design framework (Instruction, Context, Constraints, Output).
- Python 3.11 or later installed.
- An API key for OpenAI **or** Anthropic **or** a locally running Ollama model.
- Access to Claude.ai or a similar Claude interface for vibe-coding.

Task 0: Setup

> **Duration: ~15 minutes**

Five steps to get the project running before you write a single line of analysis code:

Step 1 — Copy the starter

Copy the `day4_starter_track_b/` folder to a new folder named `day4_project/`, then open that folder in your terminal:

Platform	How to copy
File manager (any OS)	Right-click `day4_starter_track_b/` → Copy, paste alongside it, rename the copy to `day4_project`
macOS / Linux terminal	`cp -r day4_starter_track_b day4_project`
Windows terminal	`xcopy /E /I day4_starter_track_b day4_project`

What you'll find in your project folder:

- `day4_project/` Root — your working copy of the project
  - `.env.example` Template — copy to .env and add your API key (Step 4)
  - `.gitignore` Keeps .env and outputs/ out of git
  - `requirements.txt` Three packages: litellm, python-dotenv, pypdf

- llm.py — Pre-provided — the LLM abstraction layer (Task 2: read it, do not modify)
- report.py — Pre-provided — renders the analysis results dict into a Markdown report file
- parse.py — You build this — Task 1
- prompts.py — You build this — Task 3
- analyzer.py — You build this — Task 4
- main.py — You build this — Task 5 (Track A) / Pre-provided (Track B)
- **inputs/** — Pre-provided sample inputs — all files already present in the starter
  - strong_resume.pdf — Pre-provided — strong résumé sample; primary input for Tasks 1–5
  - weak_resume.pdf — Pre-provided — weak résumé sample; used in Final Verification contrast runs
  - job_rtis_systems_engineer.txt — Pre-provided — the primary RTIS job description fixture
  - job_unrelated.txt — Pre-provided — unrelated job for Final Verification contrast runs
- **outputs/** — Reports land here — one .json and one .md per run
  - (empty until Task 5)

Step 2 — Create a virtual environment

```
python -m venv venv
source venv/bin/activate          # Windows: venv\Scripts\activate
```

Step 3 — Install dependencies

```
pip install -r requirements.txt
```

This installs three packages: `litellm`, `python-dotenv`, and `pypdf`.

Step 4 — Configure your API key

Duplicate `.env.example` and name the copy `.env` (both files sit in the same `day4_project/` folder). Then open `.env` in a text editor and fill in your key. The defaults work with a local Ollama model — no API key needed:

```
MODEL=ollama/gemma4:e2b
OLLAMA_API_BASE=http://localhost:11434
```

Run `ollama serve` in a separate terminal and pull the model first with `ollama pull gemma4:e2b` .

To use OpenAI instead:

```
MODEL=openai/gpt-4o-mini
OPENAI_API_KEY=sk-...
```

To use Anthropic Claude instead:

```
MODEL=anthropic/claude-3-5-sonnet-20241022
ANTHROPIC_API_KEY=sk-ant-...
```

Step 5 — Verify sample inputs

The starter pack already includes all résumé and job description files. Open the `inputs/` folder and confirm these four files are present:

- `strong_resume.pdf` — strong résumé sample (primary input for Tasks 1–5)
- `weak_resume.pdf` — weak résumé sample (used in Final Verification)
- `job_rtis_systems_engineer.txt` — RTIS job description fixture
- `job_unrelated.txt` — unrelated job description fixture

> **Note:** Both PDFs are text-based exports — `pypdf` can extract their content directly. You do not need to supply your own résumé PDF to complete this lab.

✅ Checkpoint

Run:

```
python -c "from llm import ask_json; print('LLM layer ready')"
```

What success looks like:

- The shell prints `LLM layer ready` with no traceback
- No `ModuleNotFoundError` — if one appears, re-run `pip install -r requirements.txt` inside the active virtualenv

Common errors:

Error message	Fix
`ModuleNotFoundError: No module named 'litellm'`	Run `pip install -r requirements.txt` again inside the activated virtualenv
`AuthenticationError` or `invalid_api_key`	Open `.env` and check your API key has no extra spaces
`Cannot reach Ollama at localhost:11434`	Run `ollama serve` in a separate terminal first
`Python 3.10 or lower` warning	Install Python 3.11 from python.org

Track B — Orientation Step

Before building anything, open `main.py` in your editor and find the import block near the top:

```
from parse import read_resume_pdf, read_jd_text
from analyzer import (
    extract_resume_profile, extract_jd_profile,
    analyse_keyword_match, analyse_bullets,
    analyse_jargon, analyse_structure,
    analyse_background_fit, summarise_overall,
    compute_overall_score,
)
```

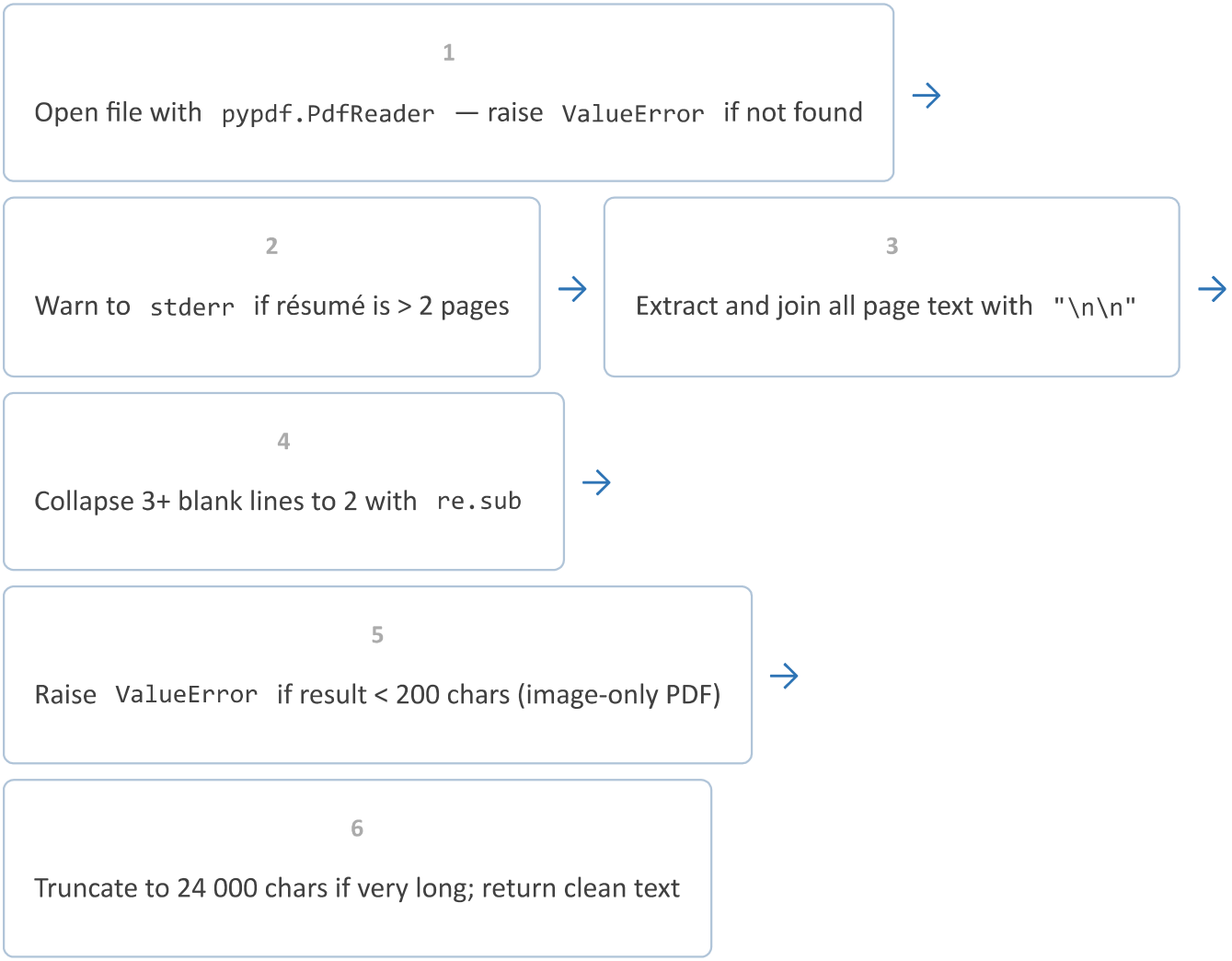
> **These are the functions you need Claude to write across the next three tasks.** `main.py` is your blueprint — it tells you exactly which names and signatures the files you create must export. The Checkpoints at each task will confirm that Claude's output satisfies these contracts.

Task 1: Parse the Résumé

> **Study material:** §5 Document Parsing

What this module does: `parse.py` reads a PDF résumé file and returns clean plain text. It also reads plain-text job description files. This stage contains **no LLM calls** — it is pure document I/O with validation. The study material explains why we validate inputs *before* passing them to the LLM (§5: the validate-before-LLM rule).

How `read_resume_pdf` processes a file — every step in order:



You need to implement two functions:

```python
def read_resume_pdf(path: str) -> str: ...

def read_jd_text(path: str) -> str: ...
```

> **Open `parse.py` in your editor.** You are **creating this file** in Task 1. The starter includes docstrings and function signatures — implement the two function bodies below.

Ask Claude to write `parse.py`

Use the prompt below as a starting point. You may adjust the wording — what matters is that you describe all the requirements clearly. The more precise your description, the better Claude's output will be.

"Write a Python file called `parse.py`. It needs two functions:

`read_resume_pdf(path: str) -> str` — uses `pypdf.PdfReader` to open a PDF file. If the file is not found or cannot be opened, it raises `ValueError` with a clear message. If the résumé has more than 2 pages, it prints a warning to `sys.stderr`. It joins all page text with `"\n\n"`, collapses runs of 3 or more blank lines to 2 using `re.sub`, raises `ValueError` if the extracted text is shorter than 200 characters (meaning the PDF is image-based), and truncates to 24 000 characters if very long (with a warning to `sys.stderr`).

`read_jd_text(path: str) -> str` — reads a UTF-8 plain text file, raises `ValueError` with a clear message if the file is not found, and raises `ValueError` if the content has fewer than 100 characters after stripping whitespace."

Why describe the error conditions specifically? The Checkpoint below tests those exact behaviours. If Claude skips the 200-character check, the Checkpoint will show you — and you can ask Claude to add it.

✅ Checkpoint

Run:

```
python -c "
from parse import read_resume_pdf
text = read_resume_pdf('inputs/strong_resume.pdf')
print(f'Extracted {len(text)} chars')
print(text[:200])
"
```

What success looks like:

- Character count above 200
- The candidate's name visible in the first 200 characters of output

Also test the error case:

```
python -c "
from parse import read_resume_pdf
read_resume_pdf('inputs/does_not_exist.pdf')
" 2>&1
```

What success looks like:

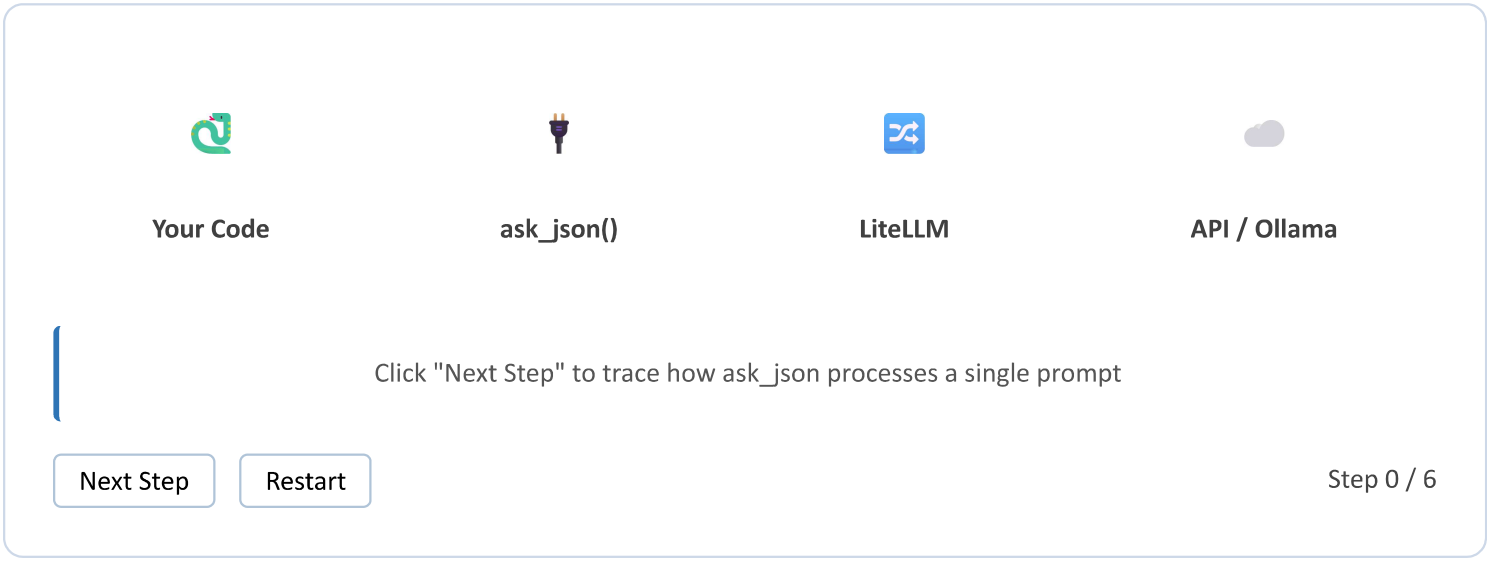
- A `ValueError` with a message like `Résumé file not found: inputs/does_not_exist.pdf`

Task 2: The LLM Layer — Read and Run

> **Study material:** §8 Handling Imperfect AI Output

What this task is: `llm.py` is pre-provided. You are **not** building it — you are reading and verifying it. You need to understand what `ask_json()` does before you use it in the next two tasks, because the quality of your prompts (and your descriptions to Claude) depends on knowing what the layer beneath them does.

Trace how `ask_json` moves a prompt through the stack:



> **Open `llm.py` in your editor.** This file is **pre-provided — do not modify it.** Read through it to understand `ask_json()` before you call it in Tasks 3 and 4.

Skim `llm.py` and answer these questions in your head before continuing:

1. *Which function removes the ```` ```json ```` code fences a model might add before its JSON response?*
2. *What is `raw_decode` doing that `json.loads` cannot? (Hint: think about what happens if the model writes "Here is the JSON:" before the actual `{...}`)*
3. *There is an anti-rewrite check near the bottom. What rule does it enforce, and why does this project need it?*

Understanding these three things will help you describe `ask_json`'s behaviour accurately when you prompt Claude to write the functions that call it.

✅ Checkpoint

Run:

```
python -c "
from llm import ask_json
result = ask_json('Return {\"status\": \"ok\"} and nothing else.', 'go')
print(result)
"
```

What success looks like:

- `{'status': 'ok'}` printed to the terminal
- One complete round-trip worked: the model returned valid JSON and `ask_json` parsed it cleanly

If you see an API error, recheck your `.env` file and API key.

Task 3: Write the Prompts

> **Study material:** §3.3 Schema-First Prompt Design · §6.1 Extraction Prompts · §6.2 Evaluation Prompts · §6.3 Feedback-Only Principle

This is the conceptual heart of the lab. Writing these system prompts applies the ICCO framework — Instruction, Context, Constraints, Output — to automated résumé analysis instead of a back-and-forth chat conversation. The

same four components structure every effective prompt here.

The eight prompts fall into three categories — each calls a different approach:

The eight prompt constants you need to write:

Constant	Type	Purpose	Key output field(s)
`RESUME_PROFILE_PROMPT`	Extraction	Candidate profile from résumé text	`name` , `skills` , `projects` , `experience`
`JD_PROFILE_PROMPT`	Extraction	Role requirements from JD text	`required_skills` , `preferred_skills`
`KEYWORD_MATCH_PROMPT`	Evaluation	Compare résumé keywords vs JD keywords	`keyword_match_score` , `missing`
`BULLET_QUALITY_PROMPT`	Evaluation	Score each bullet on ATI rubric	`bullet_quality_avg` , `bullets`
`JARGON_AUDIT_PROMPT`	Evaluation	Flag résumé/JD terminology mismatches	`jargon_score` , `flags`
`STRUCTURE_AUDIT_PROMPT`	Evaluation	Check ATS-parseability formatting	`structure_score`
`BACKGROUND_FIT_PROMPT`	Evaluation	Candidate background fit for the role	`background_fit_score`
`OVERALL_SUMMARY_PROMPT`	Synthesis	3-bullet executive summary	plain text (not JSON)

Every prompt must: - Follow ICCO structure (Instruction → Context → Constraints → Output) - Embed the exact JSON schema in the Output section (except `OVERALL_SUMMARY_PROMPT`) - End with this exact line: `"Output ONLY a valid JSON object matching the schema above. No prose. No markdown fences. No commentary. Never rewrite or generate résumé content."` - Use temperature `0.0` – `0.1` for extraction prompts, `0.2` – `0.3` for evaluation prompts

> **Open `prompts.py` in your editor.** You are **creating this file** in Task 3. Add the eight string constants described below. Each becomes a system prompt fed to `ask_json()` .

Important — no placeholders in the prompt constants. All eight constants are plain static strings — no f-strings, no `.format()` , no `{variable}` slots. The `{` and `}` characters you see inside the prompts are **literal JSON schema syntax**, not Python format placeholders. Dynamic data (the résumé text, JD text, degree code) is never embedded in the system prompt; `analyzer.py` passes it as the second argument to `ask_json()` — the user message. Your job in this task is only to write the static instruction text.

Writing prompts with Claude — worked examples

This is where schema-first prompt design (§3.3) becomes concrete. You will write two prompts in detail, then ask Claude to generate the remaining six.

Worked Example 1: `RESUME_PROFILE_PROMPT`

This prompt extracts structured information from a résumé. Use this description when prompting Claude:

"Write a Python string constant called `RESUME_PROFILE_PROMPT` for use as a system prompt. Its job: given a résumé in plain text, extract structured information and return it as a JSON object.

The JSON schema must have: - `name` (string) - `contact` (object with: email, phone, linkedin, github, portfolio — all strings) - `summary` (string — the professional summary or "" if absent) - `education` (array of objects with: school, degree, graduation_date, courses array) - `projects` (array of objects with: title, date, bullets array of strings) - `experience` (array of objects with: title, company, date, bullets array of strings) - `skills` (object with: languages, frameworks, tools, concepts, platforms — all arrays)

Key constraints: only extract what is literally present — never invent, paraphrase, or summarise. If a field is absent, return empty string or empty array. Copy bullet text verbatim.

End the prompt with exactly: 'Output ONLY a valid JSON object matching the schema above. No prose. No markdown fences. No commentary. Never rewrite or generate résumé content.'"

Worked Example 2: `KEYWORD_MATCH_PROMPT`

This compares résumé content against JD requirements. Use this description:

"Write a Python string constant called `KEYWORD_MATCH_PROMPT` for use as a system prompt. It receives a résumé profile JSON and a JD profile JSON. Its job: identify which JD keywords appear in the résumé and which are missing.

The JSON schema must have: - `present` (array of objects with: keyword, category (one of: language/framework/tool/ concept/soft_skill/buzzword), found_in (one of: summary/projects/experience/education/ skills), exact_match (boolean)) - `missing` (array of objects with: keyword, category, importance (one of: required/preferred), suggested_section, why_it_matters (string, 25 words max — diagnostic only: state what the JD says, never suggest how to change the résumé)) - `keyword_match_score` (integer 0–100, computed as: 100 × required skills found / total required skills)

Constraint: only mark a keyword as present if it can be literally located in the résumé profile fields — no inference. Even if the résumé and JD share zero keywords, both profiles are still fully provided — the model must return the schema (an empty "present" array is a valid, correct result) rather than asking for clarification or claiming no résumé was given.

End with the standard anti-rewrite line."

Generate the remaining six prompts:

Use similar descriptions for `JD_PROFILE_PROMPT`, `BULLET_QUALITY_PROMPT`, `JARGON_AUDIT_PROMPT`, `STRUCTURE_AUDIT_PROMPT`, `BACKGROUND_FIT_PROMPT`, and `OVERALL_SUMMARY_PROMPT`.

For the evaluation prompts, describe the rubric directly to Claude — these are all well-known, general conventions that need no external reference document:

- **`BULLET_QUALITY_PROMPT`** — the Action → Technology → Impact rubric (action verb + specific technology/tool + measurable impact). Describe the L1/L2/L3 reference levels and the `bullet_quality_avg` scoring formula.
- **`JARGON_AUDIT_PROMPT`** — tell Claude the model should dynamically compare résumé terminology against JD terminology and flag likely-equivalent-but-differently-worded terms (no static translation table — describe the severity rules and the `jargon_score` formula instead).
- **`STRUCTURE_AUDIT_PROMPT`** — describe general ATS-parseability rules: single-column layout, standard section headers, reverse-chronological order, appropriate length, contact info placement, no images/graphics.
- **`BACKGROUND_FIT_PROMPT`** — tell Claude the judgment must come only from `resume_profile`'s education/experience fields compared against `jd_profile`'s requirements — no external degree code or lookup table.

Why describing schemas is the hard part: This is schema-first design (§3.3) in practice. You are doing the design work; Claude is doing the typing. An imprecise description produces imprecise JSON — and the Checkpoint below will show you exactly where the gaps are.

✅ Checkpoint — Résumé Extraction

Run:

```
python -c "
from parse import read_resume_pdf
from llm import ask_json
```

```
from prompts import RESUME_PROFILE_PROMPT
text = read_resume_pdf('inputs/strong_resume.pdf')
result = ask_json(RESUME_PROFILE_PROMPT, f'RÉSUMÉ TEXT:\n\n{text}', max_tokens=2000)
print('Name:', result.get('name'))
print('Languages:', result.get('skills', {}).get('languages', []))
"
```

What success looks like:

- The candidate's real name printed
- A non-empty list of programming languages extracted from the résumé

If you see empty fields: ask Claude to revise `RESUME_PROFILE_PROMPT` — tell it specifically which fields returned empty. Provide Claude with the first 500 characters of your résumé text as evidence so it can adjust the extraction constraints.

✅ Checkpoint — Keyword Match

Run:

```
python -c "
from parse import read_resume_pdf, read_jd_text
from llm import ask_json
from prompts import RESUME_PROFILE_PROMPT, JD_PROFILE_PROMPT, KEYWORD_MATCH_PROMPT
import json
resume = read_resume_pdf('inputs/strong_resume.pdf')
jd = read_jd_text('inputs/job_rtis_systems_engineer.txt')
rp = ask_json(RESUME_PROFILE_PROMPT, f'RÉSUMÉ TEXT:\n\n{resume}', max_tokens=2000)
jp = ask_json(JD_PROFILE_PROMPT, f'JOB DESCRIPTION TEXT:\n\n{jd}', max_tokens=1500)
km = ask_json(KEYWORD_MATCH_PROMPT,
    f'RÉSUMÉ PROFILE:\n{json.dumps(rp, indent=2)}\n\nJD PROFILE:\n{json.dumps(jp, indent=2)}',
    max_tokens=3000)
print('Score:', km.get('keyword_match_score'))
print('Missing (first 3):', km.get('missing', [])[:3])
"
```

What success looks like:

- A score between 0 and 100
- A list of missing keywords (can be empty if your résumé is a strong match)

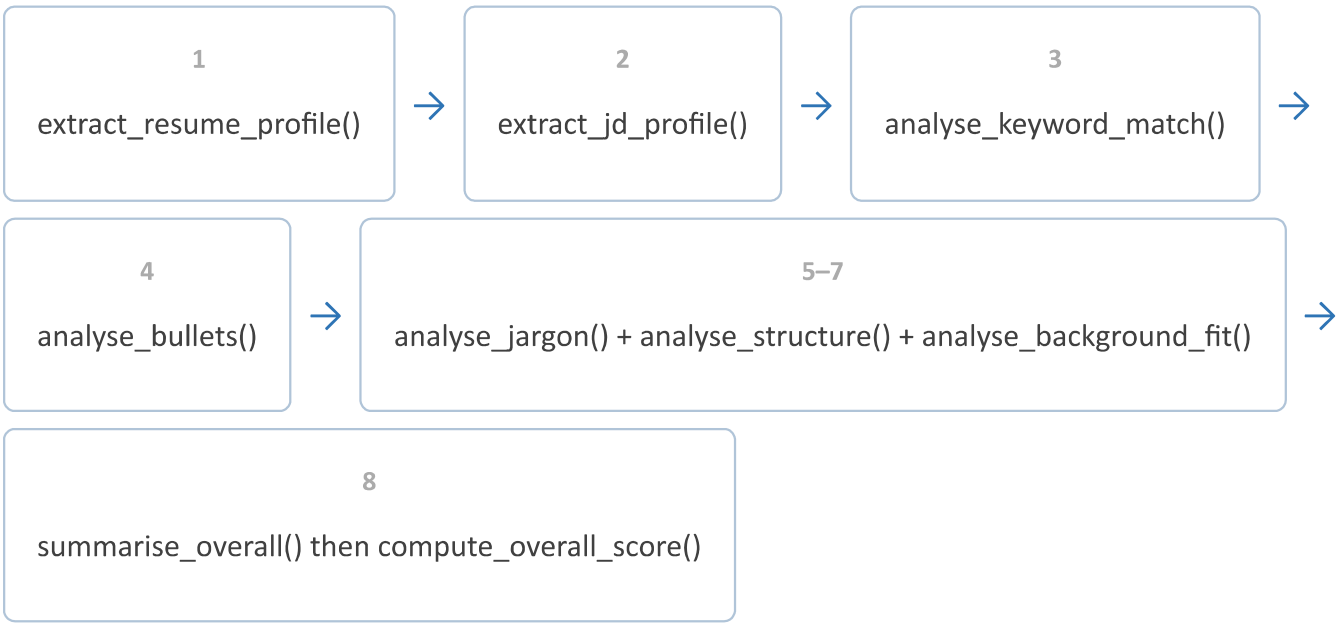
Task 4: Build the Analysis Pipeline

> **Study material:** §4 The Multi-Stage Pipeline · §7.2 Weighted Aggregation

What this module does: `analyzer.py` contains 8 analysis functions that each wrap exactly one LLM call, plus one pure-Python scoring function (`compute_overall_score`) with no LLM call at all.

The study material §4.2 says: *smoke-test each layer; integrate after each layer is verified.* This task practices that rule — the Checkpoint tests a subset of functions without requiring the rest to exist yet.

All eight functions — in the order `main.py` calls them:



> **Open `analyzer.py` in your editor.** You are **creating this file** in Task 4. Function signatures and docstrings are already present in the starter — implement each function body below.

Ask Claude to write `analyzer.py`

Give Claude these four pieces of information in a single prompt:

1. The function signatures (from `main.py` 's import block you read in Task 0):

```
extract_resume_profile(resume_text: str) -> dict
extract_jd_profile(jd_text: str) -> dict
analyse_keyword_match(resume_profile: dict, jd_profile: dict) -> dict
analyse_bullets(resume_profile: dict) -> dict
analyse_jargon(resume_profile: dict, jd_profile: dict) -> dict
```

```
analyse_structure(resume_text: str) -> dict
analyse_background_fit(resume_profile: dict, jd_profile: dict) -> dict
summarise_overall(report: dict) -> str
compute_overall_score(report: dict) -> int
```

2. How `ask_json` and `ask_text` work (from your reading of `llm.py` in Task 2): Each function calls `ask_json(system_prompt, user_message, max_tokens=N)` and returns the resulting dict. `summarise_overall` uses `ask_text(...)` and returns a plain string.

3. Which prompt constant each function uses (from the table in Task 3 above).

4. The exact scoring weights for `compute_overall_score`:

"Write `compute_overall_score(report: dict) -> int`. It reads: - `report['keyword_match']['keyword_match_score']` (weight 0.40) - `report['bullets']['bullet_quality_avg']` (weight 0.25) - `report['structure']['structure_score']` (weight 0.15) - `report['jargon']['jargon_score']` (weight 0.10) - `report['background_fit']['background_fit_score']` (weight 0.10)

Multiply each by its weight, sum them, return `int(round(total))`."

Having to describe the scoring formula forces you to engage with §7.2 — the formula is not arbitrary, and understanding the weights helps you interpret the final scores.

✅ Checkpoint

Run:

```
python -c "
from parse import read_resume_pdf, read_jd_text
from analyzer import extract_resume_profile, extract_jd_profile, analyse_keyword_match, comput
import json
resume = read_resume_pdf('inputs/strong_resume.pdf')
jd = read_jd_text('inputs/job_rtis_systems_engineer.txt')
rp = extract_resume_profile(resume)
jp = extract_jd_profile(jd)
km = analyse_keyword_match(rp, jp)
fake_report = {
    'keyword_match': km,
    'bullets': {'bullet_quality_avg': 60},
    'structure': {'structure_score': 70},
    'jargon': {'jargon_score': 90},
    'background_fit': {'background_fit_score': 80},
}
```

```
print('Composite score:', compute_overall_score(fake_report))
"
```

What success looks like:

- Terminal prints `Composite score:` followed by a number between 0 and 100 (e.g. `Composite score: 63`)
- No traceback — this exercises `extract_resume_profile`, `extract_jd_profile`, `analyse_keyword_match`, and `compute_overall_score` without requiring the remaining functions yet

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Task 5: Wire the Pipeline

> **Study material:** §4.3 Data Flow Between Stages
>
> **Open `main.py` in your editor.** This file is **pre-provided** in your Track B starter — do not modify it. This task is verification only: run the Checkpoint below and confirm the output looks correct.

Track B: The pipeline is already wired. If any `[N/8]` step raises an error, that module needs fixing — go back to the relevant task.

✅ Checkpoint

Run:

```
python main.py inputs/strong_resume.pdf inputs/job_rtis_systems_engineer.txt
```

What success looks like:

```
Using model: ollama/gemma4:e2b
[1/8] Parsing résumé: inputs/strong_resume.pdf
[2/8] Reading JD: inputs/job_rtis_systems_engineer.txt
[3/8] Extracting résumé profile (LLM)...
[4/8] Extracting JD profile (LLM)...
[5/8] Keyword match (LLM)...
[6/8] Bullet audit (LLM)...
[7/8] Jargon, structure, background fit (LLM x3)...
[8/8] Final summary (LLM)...

Score: XX/100  (PASS 60% ATS threshold)
```

```
JSON:  outputs/match_report_YYYYMMDD_HHMMSS.json
MD:    outputs/match_report_YYYYMMDD_HHMMSS.md
```

- `<executive summary bullet 1>`
- `<executive summary bullet 2>`
- `<executive summary bullet 3>`

Score below 60 shows `FAIL` instead of `PASS`. Two report files should appear in `outputs/`.

Final Verification — All Four Combinations

You have built a working AI analysis pipeline. These four runs are a designed experiment: each combination holds one variable constant while changing the other. Your goal is not to hit a score target — it is to watch your tool produce different outputs from the same code and understand *why*.

There is no grading and no submission. Open the `.md` report from `outputs/` after each run and read what your system produced.

```
# Run 1 — Best case: strong résumé, relevant job
python main.py inputs/strong_resume.pdf inputs/job_rtis_systems_engineer.txt

# Run 2 — Weak résumé, same relevant job (isolates résumé quality)
python main.py inputs/weak_resume.pdf inputs/job_rtis_systems_engineer.txt

# Run 3 — Strong résumé, wrong job (isolates job fit)
python main.py inputs/strong_resume.pdf inputs/job_unrelated.txt

# Run 4 — Worst case: both variables fail
python main.py inputs/weak_resume.pdf inputs/job_unrelated.txt
```

After each run, open the `.md` report and check the **Score Breakdown** table at the bottom — it shows exactly which of the five components pulled the total up or down.

Run	Expected verdict	What to look for in the `.md` report
1: strong + relevant job	PASS (65+)	High `keyword_match_score`; long "present" keyword list; few or no terminology flags
2: weak + relevant job	FAIL (35–55)	Long `missing` keyword list with many `required` items; Bullet Quality Audit shows mostly L1/L2; compare Score Breakdown with Run 1 row by row
3: strong + unrelated	FAIL (20–45)	`keyword_match_score` collapses; `bullet_quality_avg` is nearly identical to Run 1 — only one thing changed
4: weak + unrelated	FAIL (10–30)	Every Score Breakdown row is low; this is the floor — use it as a reference when reading Runs 1–3

Reflection

These are not graded questions. They are prompts for your own reasoning — worth thinking through now, before you extend this project further.

1. Quality versus relevance

Place the Score Breakdown tables from Run 1 and Run 3 side by side (open both `.md` reports). The résumé is identical in both runs.

- Which rows in the breakdown changed between the two?
- Which rows stayed the same?
- The score difference between Run 1 and Run 3 is driven almost entirely by one component. Which one, and what does that tell you about what an ATS actually measures?

2. What the weak résumé is missing

Open the Run 2 `.md` report (weak résumé + relevant job). In the Keyword Match section, find the first three `required` keywords listed under "missing."

- Are those skills the candidate never learned — or skills they likely have but did not name in the résumé?
- If it is the latter, what does that imply about the value of keyword visibility versus actual skill level?

3. Inside a bullet

Open the Bullet Quality Audit in any run. Find one bullet rated L1 or L2. Read its `what_is_missing` field.

- Which of the three ingredients (Action, Technology, Impact) is absent?
- Without rewriting the bullet text, describe in one sentence what specific information would push it to L3.

4. The 40% weight

`keyword_match_score` carries 40% of the total — the largest single weight. Your Run 3 shows a strong résumé failing because of keyword mismatch alone.

- Why might a real ATS weight keyword density so heavily relative to writing quality?
- Is that weighting fair to a candidate with strong work who is switching industries?
- If you were redesigning the weights for a design-focused role (e.g. UX/UI design) rather than a systems-engineering role, what would you change?

What You Built

Your pipeline ran on all four combinations and produced eight report files. You implemented:

- A PDF parser that validates inputs before any LLM call
- Eight prompt constants grounding evaluation in well-known, general résumé/ATS and hiring conventions
- An eight-call LLM pipeline with structured JSON output at every stage
- A weighted scoring formula and a human-readable Markdown report

The `.md` reports in `outputs/` are real, grounded screening feedback — produced by the system you wrote, in the same shape a company's own applicant-screening pipeline would produce.